

Drone Telemetry Simulation – Project Overview

1. What I Built

I developed a Drone Telemetry Simulation System using Python. This program simulates real-time drone flight data such as altitude, speed, battery level, and GPS coordinates over a 30-second flight period. The system mimics how an actual drone continuously sends telemetry data to a ground control station for monitoring and analysis.

2. What the System Does

- Altitude (meters) – Simulates vertical movement.
- Speed (m/s) – Simulates velocity changes.
- Battery (%) – Gradually decreases to simulate power consumption.
- GPS Coordinates (Latitude & Longitude) – Simulates drone movement across a geographic area.
- Time Elapsed (seconds) – Tracks simulation duration.

The program runs for 30 seconds, updating and printing new telemetry data once every second to simulate real-time flight monitoring.

3. How It Works

Initialization

Altitude	100 meters
Speed	10 m/s
Battery	100%
Initial GPS Location	(17.3850, 78.4867)

Real-Time Simulation Loop

- Altitude Change: `altitude += random.randint(-2, 5)` — Simulates climb or descent.
- Speed Change: `speed += random.randint(-1, 2)` — Simulates acceleration or deceleration.
- Battery Drain: `battery -= random.uniform(0.5, 1.5)` — Simulates realistic battery consumption.
- GPS Movement: `latitude/longitude += random.uniform(-0.0005, 0.0005)` — Simulates small geographical movement.
- Output Display: Updated telemetry values are printed in a structured format.
- Real-Time Delay: `time.sleep(1)` — Ensures a 1-second delay between updates.

4. Tools and Software Used

- Programming Language: Python

- Libraries Used: random (for realistic variations), time (for real-time delay)
- Development Environment: Any Python IDE or terminal (VS Code, PyCharm, Jupyter Notebook, or Command Prompt)

5. Key Features

- Real-time telemetry simulation
- Randomized but realistic drone behavior
- Battery safety check (prevents negative values)
- Continuous monitoring for 30 seconds
- Simple and readable console output

6. Conclusion

This project demonstrates a basic but realistic implementation of a drone telemetry system using Python. It simulates how drones transmit live flight data and provides a foundation for further expansion, such as connecting to a graphical dashboard, adding low-battery alerts, exporting data to a file, and integrating with real drone hardware. The simulation successfully models core telemetry concepts used in real-world drone monitoring systems.